EE 582: Survey **November 26, 2025**

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Please answer the following questions:

Q1: The term is almost over, and we learned many things in this course. Which topics have been covered? (check the box ✓)

- ☑ Trends of energy systems, and applications of power electronic converters and electric machines in various fields and industries
- ☑ Purpose of modeling the energy systems, nature of time and concept of discretization, various types of models, time-scale of phenomena in power systems, various simulation tools for power system studies and their methods of solving the network
- ☑ State-space modeling of electric circuits, various integration methods and ODE solvers, explicit and implicit solvers, multi-step methods, numerical stability and stiffness, variable-step ODE solvers and simulation step-size control, Simulink ODE solvers and how to choose them
- ☑ Nodal-analysis-based modeling of electric circuits for electromagnetic transient (EMT) simulations, discretization of circuit components with various integration rules, companion circuits and nodal network solutions, modeling switching devices, modeling nonlinear components, network solution with Gaussian elimination and LU decomposition, simulation step-size selection for nodal-based EMT simulations, accuracy and numerical stability of various integration rules, numerical oscillation and damping adjustments, modeling transmission lines, discrete-time modeling of transmission lines based on traveling waves
- ☑ Theory of reference frame transformation, stationary and rotating reference frames, $\alpha\beta$ and qd coordinates, modeling of electric circuits in arbitrary reference frames, transformation between the reference frames
- ☑ Types and construction of commonly used induction machines and their application, steady-state equivalent circuits, power conversion and torque-speed characteristics, principles of speed and torque control, dynamic modeling of induction machines for transient analysis, various types of dynamic models (coupled-circuit phase domain and $qd0$ models), implementation of these models in Simulink
- ☑ Types and construction of commonly used synchronous machines and their application, steady-state equivalent circuits, power conversion and torque-speed characteristics, dynamic modeling of synchronous machines for transient analysis, various types of dynamic models (coupled-circuit phase domain and $qd0$ models), per-unit system, implementation of these models in Simulink
- ☑ Various types of power-electronic converters, full-bridge inverters and power quality analysis, amplitude and harmonic control, various pulse-width modulation (PWM) techniques, six-step three-phase inverters, three-phase PWM-controlled inverters, sinusoidal PWM (SPWM) and space-vector modulation (SVM) schemes, average-value models of switching converters, multi-level inverters
- ☑ Different methods of controlling grid-connected converters, modeling and control in $\alpha\beta$ and qd frames, controlling active and reactive power, synchronization with the ac grid and PLL design, dc-side voltage control, grid-forming converters (virtual synchronous generator, droop-based control, etc.), load voltage control, autonomous (for islanding) operation
- Later ☐ Introduction to real-time simulators (study material will be provided)

Q2: Wow! Based on **Q1**, it seems that we have covered a lot of materials that are very important in Power Engineering, especially with the growth of the need for expertise in simulating energy systems. I agree that there were so many and the load may have been pretty high, but also believe that learning these important concepts will pay-off when you enter industry (soon!). What is your opinion?

- ☑ Agree.
- ☐ Partly agree. Please explain a bit.
- ☐ Disagree. Please explain.

Q3: Which topic(s)/module(s) was (were) the <u>most</u> interesting to you?

Nodal Analysis, State Space Analysis and Travelling Wave, All Power Electronics, GFM and GFL Converters

Q4: Which topic(s)/module(s) was (were) the <u>least</u> interesting to you?

Induction machines, Synchronous machines - I'm personally not very interested in machines, but I understand that they are essential topics.

Q5: Which topic(s)/module(s) do you think should be shortened and paid less attention? or were they all good?!

All topics are esssential and well covered, but I feel the content was too much for a single 3 credit course. So maybe we spend less time on the machine topics and students are encouraged to self-study and implement either induction or synchronous machines (like the case with the project) by themselves.

Q6: Which topic(s)/module(s) do you think should be covered in more detail? or were they all good?!

No topic was covered in less than ideal detail I feel

Q7: I tried my best during the semester to deliver the course professionally with commitment to covering all the tentative topics! I also tried to provide you with all the possible material (video recordings of the lectures, lecture notes, my own annotations, simulation files and demos, reference papers, etc.).

Overall, are you satisfied with the teaching? (Your feedback is appreciated!)

- ☑ Yes (Extremely)
- ☐ Partly. Please explain a bit.
- ☐ No. Please explain.

<u>To remember:</u> You will get 10% points on this survey! Thanks for providing feedback! The last assignment (HW6) will also be as 10% bonus! Also, your lowest homework grade will be dropped. ☺

Cheers!

I enjoyed the semester with You! ☺

Wish you the best of luck with your finals! ☺

Happy Thanksgiving!